Homework 12
Introduction to Complex Analysis
(MATH 4230-40), Spring 2022

Section 4.4 - Evaluation of Infinite Series and Partial-Fraction Expansions

1. Show that $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{(2n-1)^3}=\frac{\pi^3}{32}$. You may assume that $\displaystyle\lim_{N\to\infty}\int_{C_N}\left(\frac{\pi}{\sin(\pi z)}\right)\left(\frac{1}{(2n-1)^3}\right)dz=0$, where C_N is the square with corners $\pm(N+1/3)\pm(N+1/3)i$.

Note that $G(z)=\dfrac{\pi}{\sin(\pi z)}$ has a simple pole at each integer and $\operatorname{Res}(G(z);n)=(-1)^n$ Let us take $f(z)=\dfrac{1}{(2z-1)^3}$ so that $f(n)=\dfrac{1}{(2n-1)^3}$ and note that f has a pole of order 3 at $z=\dfrac{1}{2}$. So,

$$\int_{C_N} G(z)f(z)dz=2\pi i\sum_{-N}^{N}(-1)^n f(n)+2\pi i\sum\{\text{residues of } G(z)f(z) \text{ at residues of } f(z) \text{ in } C_N.\}$$

By assumption, $\displaystyle\lim_{N\to\infty}\int_{C_N}G(z)f(z)dz=\lim_{N\to\infty}\int_{C_N}\left(\frac{\pi}{\sin(\pi z)}\right)\left(\frac{1}{(2n-1)^3}\right)dz=0$, so this becomes

$$\sum_{-\infty}^{\infty}\frac{(-1)^{n-1}}{(2n-1)^3}=\sum\{\text{residues of } G(z)f(z) \text{ at residues of } f(z).\}=\operatorname{Res}\big(f(z)G(z);1/2\big).$$

Observe that

$$\begin{aligned}
\sum_{n=-N}^{N}\frac{(-1)^{n-1}}{(2n-1)^3}&=\sum_{n=-N}^{0}\frac{(-1)^{n-1}}{(2n-1)^3}+\sum_{n=1}^{N}\frac{(-1)^{n-1}}{(2n-1)^3}\\
&=\sum_{n=0}^{N}\frac{(-1)^{(-n)-1}}{(2(-n)-1)^3}+\sum_{n=1}^{N}\frac{(-1)^{n-1}}{(2n-1)^3}\\
&=\sum_{n=0}^{N}\frac{(-1)^{n}}{(2n+1)^3}+\sum_{n=1}^{N}\frac{(-1)^{n-1}}{(2n-1)^3}\\
&=\sum_{k=1}^{N+1}\frac{(-1)^{k-1}}{(2(k-1)+1)^3}+\sum_{n=1}^{N}\frac{(-1)^{n-1}}{(2n-1)^3}\\
&=\sum_{k=1}^{N}\frac{(-1)^{k-1}}{(2k-1)^3}+\frac{(-1)^N}{(2N-1)^3}+\sum_{n=1}^{N}\frac{(-1)^{n-1}}{(2n-1)^3}=\frac{(-1)^{N-1}}{(2N-1)^3}+2\sum_{n=1}^{N}\frac{(-1)^{n-1}}{(2n-1)^3}.
\end{aligned}$$

As $N\to\infty$, it is easy to see that $\dfrac{(-1)^{N-1}}{(2N-1)^3}\to 0$. So, we have $\displaystyle\frac{1}{2}\sum_{n=-\infty}^{\infty}\frac{(-1)^{n-1}}{(2n-1)^3}=\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{(2n-1)^3}$. Doing some computation to obtain $\operatorname{Res}\big(f(z)G(z);1/2\big)=\dfrac{\pi^3}{16}$, we produce $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{(2n-1)^3}=\frac{1}{2}\left(\frac{\pi^3}{16}\right)=\frac{\pi^3}{32}$.

Section 5.1 - Basic Theory of Conformal Mappings

2. Near what points are the following maps conformal?

   (a) $f(z) = \overline{z}$

   A mapping $f : A \to \mathbb{C}$ is conformal at a point $z_0 \in A$ if f is analytic at z_0 and $f'(z_0) \neq 0$. Let us write $f(x+iy) = u(x,y) + iv(x,y)$ so that $u(x,y) = x$ and $v(x,y) = -y$. Then, we have

   $$\frac{\partial u}{\partial x} = 1 \qquad \frac{\partial v}{\partial x} = 0$$
   $$\frac{\partial u}{\partial y} = 0 \qquad \frac{\partial v}{\partial y} = -1$$

   So, while both u and v are continuously differentiable (in the sense of real variables), $\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial u}$. So, the Cauchy-Riemann equations are not satisfied and, by Theorem 1.5.8, f is not analytic at any point $z \in \mathbb{C}$. Therefore, f is not conformal.

   (b) $f(z) = \dfrac{\sin z}{\cos z}$

   A mapping $f : A \to \mathbb{C}$ is conformal at a point $z_0 \in A$ if f is analytic at z_0 and $f'(z_0) \neq 0$. Observe that $f(z) = \dfrac{\sin z}{\cos z}$ is analytic for all z such that $\cos z \neq 0$, and $f'(z) = \dfrac{\cos^2 z + \sin^2 z}{\cos^2 z} = \sec^2 z$ is nonzero for all z such that $\cos z \neq 0$. Therefore, it follows that $f(z)$ is conformal on $\mathbb{C} \setminus \left\{\frac{\pi}{2} + k\pi : k \in \mathbb{Z}\right\}$.

3. If $f : A \to B$ is bijective and analytic with an analytic inverse, prove that f is conformal.

   Since $f : A \to B$ is bijective and analytic with an analytic inverse, it follows that $f^{-1}\big(f(z)\big) = z$ for all $z \in A$ and (using the chain rule) that $\left(f^{-1}\right)'\big(f(z)\big)f'(z) = 1$. So, $f'(z) \neq 0$ for all $z \in A$, since the contrary would contradict this. By the Inverse Function Theorem (i.e. Theorem 1.5.10), it follows that $\left(f^{-1}\right)'(z) = \dfrac{1}{f'(z)} \neq 0$ for all $z \in A$. That is, f^{-1} is analytic on A with a nonzero derivative on A, and thereby conformal.